Design, Sampling, and Collection

Session 1

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Reading

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

Deaton is open access (CC BY 3.0 IGO). On the hub it is already in your `reading/` folder; otherwise PDF.

The GLSS7 report is published by the Ghana Statistical Service and is not ours to redistribute: GSS catalogue 97.

Brief description of LSMS Surveys

A timeline of the surveys `lsms_library` knows about: one row per country, one bar per survey wave, so we can see overall temporal coverage — 37 countries, 113 waves, 1985 to the present.

LSMS_Library is a work in progress

Open source; see `https://github.com/ligon/lsms_library`. The idea is to provide a software *abstraction layer* to interface with LSMS-style surveys, providing a set of common labels, tables, and variables.

Note that this is not "harmonization"; we preserve all the detail of the individual surveys. Aggregation across surveys where necessary is left to the analyst.

If you find a problem, something's missing, or have a feature request: `https://github.com/ligon/lsms_library/issues`. I'd really appreciate your help!

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Survey Design

Any household survey has:

Population What is the *population* that is supposed to be represented?

Sample A collection of selected households

Sample Frame What is the practical information we have on the population that we can use to construct the sample?

Sample Weights Proportional to the *ex ante* probability that a given sample household was selected for inclusion in the sample.

Sample Data Actual information collected from actual households in the sample

Survey Design Example

Consider the Ghana Living Standards Survey (GhanaLSS), and specifically the most recent publicly released wave 7 (2016–17).

Sample Frame A list of private dwellings from the 2010 census. What does this exclude?

Sample Weights Proportional to the inverse of the *ex ante* probability that a given sample household was selected for inclusion in the sample. The weights in `lsms_library` are normalized to have a mean of one, but sometimes instead they're normalized to sum to the estimated size of the population.

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Choosing a Population

In designing a population survey, one has to first choose a population. We'll be focused on *household* surveys within a particular country.

That sounds like a pretty good start: for example, all households in Ghana. But almost immediately edge cases turn up.

- First, what's a household? Usual definition: a set of people who "share a cooking pot"
- What about people in Ghana who don't live in households?
  - Convicts who live in prisons
  - Students who live in dormitories
  - Nuns who live in monasteries

l Solution "Private households" (so we're excluding the above)

Choosing a Survey Frame

If we've settled on "private households", we have to figure out a way to actually somehow enumerate all the private households in a country so we can make a random selection.

- Possibilities
  - Use a census (the usual LSMS approach)
  - Use remote sensing and machine learning to try to identify private dwellings (Togo a recent example)
- Problems
  - Census is usually out of date
  - Remote sensing is highly error prone
- Leads to demand for ground-truthing

Census approach to enumerating households

Typical approach:

1. Divide entire country into "Enumeration Areas" (EAs).
2. Send a team to each EA; have them identify/map all residential structures (separate dwellings, apartment blocks, compounds).
3. Find someone who lives in each structure; if it's a private dwelling, list all who live there.

Two-Stage Cluster Based Sampling

At the time a census is conducted, it's usually the most reliable "ground-truth"; it provides a list of private households, you could just randomly select. But:

- Census out of date almost immediately; new construction, people move, etc.

Stages

1. Take a random sample of EAs. Then, *just for these sampled areas*, send a team to update the census listing, and get an up-to-date list of all households in just the *selected* EAs.
2. For each sampled EA, randomly sample the households within it.

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Sampling Weights

We can always compute statistics (mean, variance, etc.) that are valid for the *sample*; but usually we'd like to use the sample to say something about the *population*. For this we need to know the probability π_i that a particular sample household i was included in the sample *ex ante*.

Once we know these we can compute statistics from a sample of size N.

For the sample:

Mean $\bar{x} = \frac{1}{N}\sum_{i\in\text{Sample}} x_i$

Variance $\text{Var}(x) = \frac{1}{N}\sum_{i\in\text{Sample}} x_i^2 - \bar{x}^2$

Sampling Weights

For the population

Let $\hat{\hat{X}} = \frac{\sum_{i \in \text{Sample}} x_i/\pi_i}{\sum_{i \in \text{Sample}} 1/\pi_i}$ (this is the Hájek estimator).

Mean $\bar{X} = \mathbb{E}\left[\hat{\hat{X}}\right] + O(1/N)$

Variance $\mathbb{E}(X - \bar{X})^2 = S^2 =$

$$\mathbb{E}\left[\frac{\sum_{i \in \text{Sample}} (x_i - \hat{\hat{X}})^2/\pi_i}{\sum_{i \in \text{Sample}} 1/\pi_i}\right] + O(1/N)$$

Takeaway: To draw inferences about the population we need to know sampling probabilities π_i. *It is not enough to just have the data!*

Sub-Populations

Conducting a large-scale household survey is expensive! Sometimes you'd like to draw inferences not just about the original population (e.g., all the private households in Ghana), but also for subpopulations: e.g.,

- Particular regions
- Rural vs. Urban

The GLSS: Later rounds of the GhanaLSS are designed to do *both*; for example, to make population statements about the urban population of the Ashanti region.

Stratification

The most statistically efficient way to do inference for sub-populations is via stratification: Partition the overall population into different parts, and (effectively) treat each one as it's own sampling problem.

- Let N_s be the size of the sample for stratum s, and n_s be the size of the population in the stratum.

Strata Weights

If $N_s/n_s \neq N/n$ (the sample drawn in stratum s isn't proportional to the share of s in the population) then we've violated the maxim that every household in the overall population has an equal chance of being selected. To correct this, construct strata weights

$$v_s = \frac{N_s/n_s}{N/n}$$

to correct for the fact that a household in stratum s has a probability of being selected into the sample which must be adjusted by the factor v_s.

Two-Stage Sampling Weights

In a Two-Stage Clustered sampling scheme, the probability of a particular household i who lives in EA c_i being included in the sample depends on two other probabilities:

- The probability of EA (cluster) c_i being selected in the first stage, say p_{c_i};
- *Conditional* on c_i being selected, the probability of i being selected from all the other households in the cluster, say r_i.

So, in a two stage clustering scheme:

$$\pi_i = p_{c_i} r_i$$

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Inferring sample design from weights

For household i in cluster c_i, a two-stage design factors the probability of selection:

$$\pi_i = \underbrace{p_{c_i}}_{\text{cluster chosen}} \times \underbrace{\frac{b_{c_i}}{M_{c_i}}}_{r_i\text{: chosen within it}}$$

b_{c_i} is the number of households in the cluster selected (twelve); M_{c_i} is the size of the cluster. The weight is the reciprocal:

$$w_i \;\propto\; \frac{1}{\pi_i} \;=\; \frac{M_i}{p_i\, b_i}$$

One equation, two unknowns. We see w and b; we want p and M.

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